

LTE C-V2X Vehicular Communication Simulation & Performance Evaluation

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Abstract— This paper evaluates the performance of a C-V2X application using LTE Mode 4 in Veins/OMNeT++, focusing on **Packet Reception Ratio (PRR)** in downtown and suburban Montreal. As expected, increased vehicle density significantly reduced PRR under both Nakagami and Jakes fading, as shown in figure 1. However, contrary to the initial intuition of improved performance in a less obstructed environment, the suburban area exhibited a slightly *lower* PRR compared to the downtown area, suggesting that factors beyond static obstacle density may be playing a more significant role in communication reliability under the simulated conditions.

INTRODUCTION

To assess the performance of a Cellular Vehicle-to-Everything (C-V2X) communication application under varying environmental conditions, two distinct scenarios were simulated: (1) a high-obstacle-density environment modeled after downtown Montreal, and (2) a lower-obstruction suburban setting represented by LaSalle, Montreal. The evaluation focuses on Packet Reception Ratio (PRR), a key metric for assessing the reliability of vehicular communications.

The wireless channel simulation employed established fading models, specifically Nakagami and Jakes, to emulate realistic signal propagation in both the urban and the suburban environments. These models allow for the examination of the impact of both multipath and mobility-induced fading. The results of this study contribute to understanding the interplay between environmental complexity, vehicular density, and the characteristics of wireless fading on the reliability of C-V2X sidelink communication.

CALCULATING PACKET RECEPTION RATIO (PRR)

The Packet Reception Ratio was calculated on a per-vehicle basis by dividing the total number of packets received at the MAC layer from all other vehicles by the total number of packets transmitted by that specific vehicle. These values were extracted from the simulation outputs, specifically the scalar metrics:

- sentPacketToLowerLayer:count (transmissions)
- receivedPacketFromLowerLayer:count (receptions)

Only values corresponding to the .mac module were used to ensure accurate MAC-layer analysis. The simulation data was processed using Python, which calculated PRR per vehicle and computed the scenario-wide average PRR. The code used for this analysis is included in the appendix.

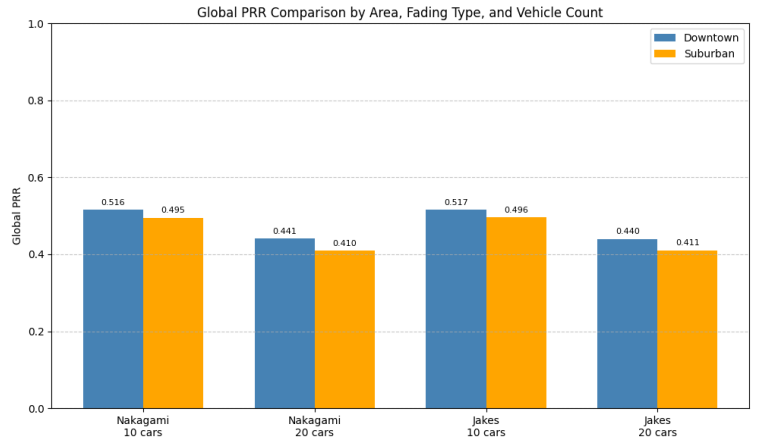


Figure 1: Global average PRR Comparison by Area, Fading Type, and Vehicle Count

CHANNEL CONFIGURATIONS & FADING MODEL

The wireless channel characteristics for the simulations were configured using the config-channel.xml file within the OMNeT++ simulation environment. This file specifies several key parameters that influence signal propagation and interference:

- **Channel Model:** The channel model was set to "REAL," indicating that the simulations aimed to emulate realistic radio wave propagation rather than using a simplified, abstract model.
- **Interference Settings:** Inter-cell interference computation and interference from other UEs were enabled, while the multi-cell interference was disabled.
- **Dynamic-LoS:** is disabled, this is a significant simplification, likely affects the accuracy of the results, particularly in the downtown scenario, where NLoS conditions are prevalent, leading to overestimation in

PRR. Trying to enable it, the simulation terminated due to severe signal loss.

- **Fixed LoS:** this setting is set to true, indicating that the simulation will always compute path loss using the LoS path loss model, regardless of whether an actual obstruction (like a building) exists between the transmitting and receiving vehicles.

To evaluate the impact of channel fading on communication reliability, simulations were conducted using three fading models: Nakagami, Rayleigh, and Jakes.

1. Nakagami channel model

The Nakagami model was used to simulate random fluctuations in signal strength due to multipath propagation. The severity of fading is controlled by the m parameter, where values less than 1 represent severe fading, and values greater than 1 model less severe conditions. When $m = 1$, the Nakagami model reduces to the Rayleigh model.

2. Rayleigh channel model

Attempting to simulate Rayleigh fading resulted in simulation failure after only a few vehicles due to extreme signal degradation. This indicated that Rayleigh fading, under the current setup, imposed unrealistically harsh conditions for LTE Mode 4 communication in this scenario.

3. Jakes channel model

The Jakes model, unlike Nakagami or Rayleigh, is a deterministic model that emulates time-varying channel fading due to mobility, particularly through Doppler shift. In this study, the Jakes model was configured with six sinusoidal paths ($path\ count = 6$) to simulate the Doppler spectrum. A higher path count yields smoother and more accurate fading behavior, better capturing mobility-induced signal variation.

RESULTS & DISCUSSION

I. THE EFFECT OF THE FADING MODEL

The Simulation results obtained using the *Jakes model* showed behavior closely aligned with Nakagami fading, supporting the idea that Doppler-induced fading dominates in this mobile vehicular context. The relative consistency between Jakes and Nakagami results implies that *static multipath fading plays a limited role*, and the primary challenge to link reliability stems from movement-related variations in signal strength.

Table 1: PRR difference between Nakagami and Jakes fading models

Area	No. Vehicles	PRR Nakagami	PRR Jakes	%from Nakagami to Jakes
Downtown	10	0.516	0.513	-0.58%
Downtown	20	0.411	0.440	-0.23%
Suburban	10	0.490	0.496	0.20%
Suburban	20	0.410	0.411	0.24%

From table1, in suburban area the performance evaluation using Jakes fading model is usually more than the

performance evaluation using a Nakagami model. This is likely due to the smaller impact of the static multipath represented by Nakagami, compared to the more dominant effect of the doppler effect that causes decrease in PRR when considered in Jakes model.

II. THE IMPACT OF VEHICLE DENSITY

The results demonstrate a noticeable decline in PRR as the number of vehicles increases from 10 to 20 under both Nakagami and Jakes fading models, as in *Figure 1*. As expected, increasing the number of vehicles increases also the channel congestion and results in more packets collisions and subsequent drops.

The results are summarized in table 2, for the decrease in PRR because of the increasing vehicular density from 10 to 20.

Table 2: Decrease in PRR due to increasing the number of vehicles

Area	Fading Model	10 Vehicles	20 Vehicles	PRR Difference
Downtown	Nakagami	0.516	0.441	-14.15%
Downtown	Jakes	0.513	0.440	-14.23%
Suburban	Nakagami	0.490	0.410	-16.13%
Suburban	Jakes	0.496	0.411	-17.14%

Figure 2 and Figure 3 show the results in an urban downtown area of Montreal, the global PRR under Nakagami fading from 0.516 (10 cars) to 0.441 (20 cars), a 14.5% decrease. While under Jakes fading, it decreases from 0.517 to 0.440, corresponding to a 14.9% decrease.

Figure 4 and Figure 5 show the results in a suburban environment, increasing the number of vehicles from 10 to 20 also leads to a noticeable decrease in the PRR, like the trend observed in the downtown area.

Under Nakagami fading, the global PRR decreases from 0.495 with 10 cars to 0.410 with 20 cars, representing a 17.2% decrease. A similar trend is evident with the Jakes model, where the PRR drops from 0.496 to 0.411, corresponding to a 17.1% decrease.

Additionally, the individual PRRs show more variability among cars, suggesting that *not all vehicles are equally affected*, likely due to factors including specific traffic flow patterns simulated by SUMO, making one car face more severe channel conditions like shadowing or greater interference from other UEs.

A noticeable example is *car_12* in figure 5, the PRR is 0.04 likely due to the larger inter-distance between the vehicle and its neighbors.

Overall, the data confirms that increased vehicle density leads to reduced overall communication reliability represented by PRR.

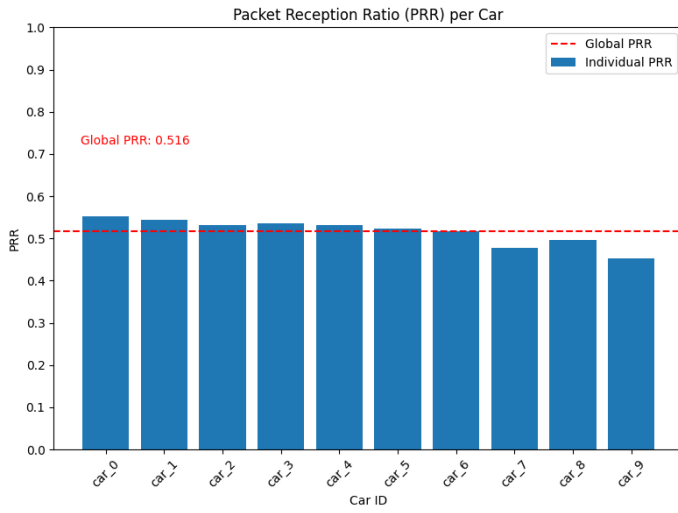


Figure 2: PRR in a **downtown Montreal area**, Nakagami fading model, for Vehicles density of 10

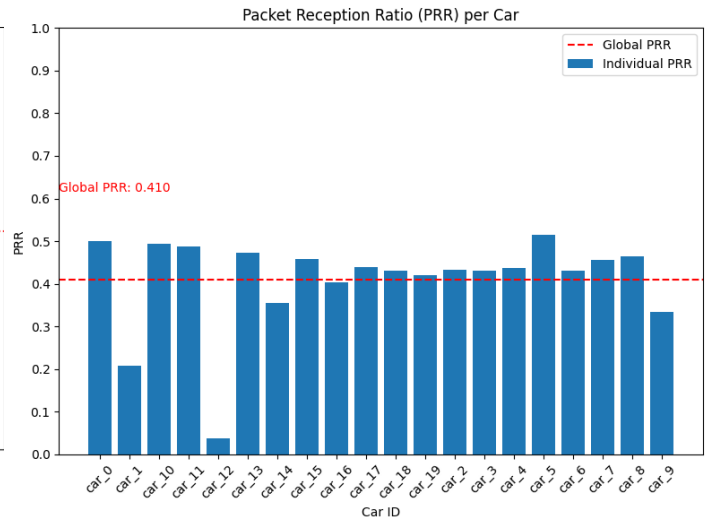


Figure 5: PRR in a **suburban Montreal area**, Nakagami fading model, for Vehicles density of 20

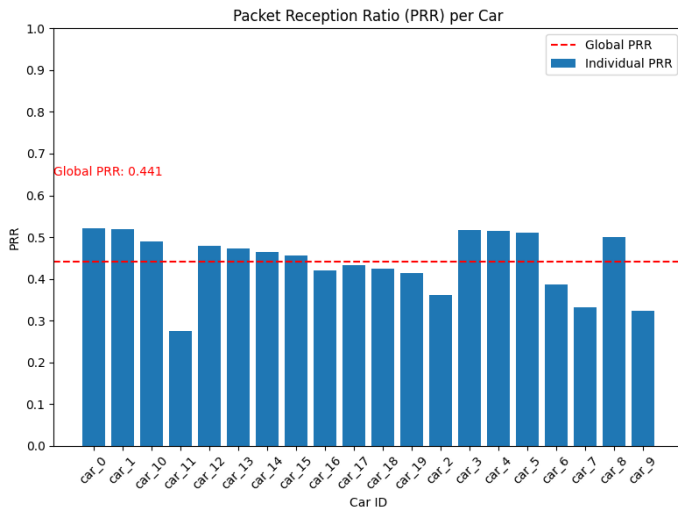


Figure 3: PRR in a **downtown Montreal area**, Nakagami fading model, for Vehicles density of 20

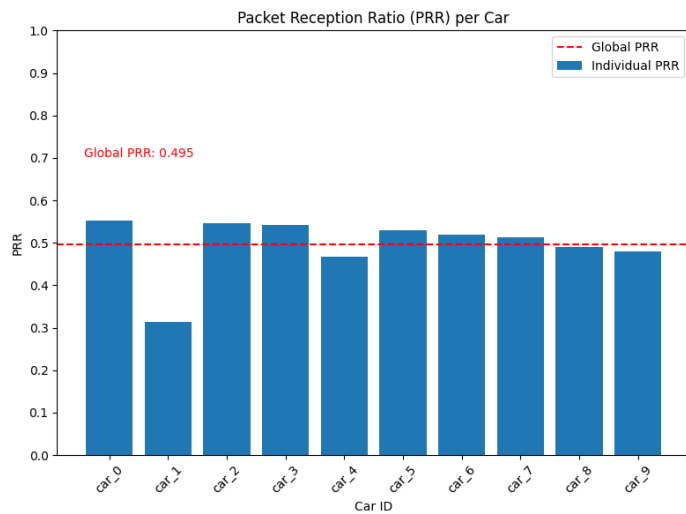


Figure 4: PRR in a **suburban Montreal area**, Nakagami fading model, for Vehicles density of 10

III. THE IMPACT OF OBSTACLE-DENSITY ENVIRONMENT

Despite the significant differences in physical obstruction between the downtown and suburban environments, the difference in PRR is relatively small.

For instance, under the Nakagami fading model with 10 vehicles, the global PRR is 0.516 in the downtown area and 0.495 in the suburban area, a difference of only 4.1%.

When the number of vehicles is increased to 20, the global PRRs remain close—0.441 (downtown) vs. 0.410 (suburban) – a difference of 7.0%.

Under the Jakes fading model with 10 vehicles, the global PRR is 0.517 in the downtown area and 0.496 in the suburban area, a difference of 4.06%.

Even when the number of vehicles is increased to 20, the global PRRs remain close—0.440 (downtown) vs. 0.411 (suburban) – a difference of 6.59%.

These results are summarized in table 3.

Table 3: PRR difference between downtown and suburban

Area	No. Vehicles	Downtown	Suburban	PRR Difference
Nakagami	10	0.516	0.495	-5.04%
Nakagami	20	0.441	0.410	-6.80%
Jakes	10	0.517	0.496	-3.31%
Jakes	20	0.440	0.411	-6.82%

This suggests that mobility-related factors, such as Doppler shifts and vehicle density, may exert a stronger influence on PRR than static environmental characteristics alone. However, it's important to acknowledge that this simulation used a uniform building height across both the downtown and suburban scenarios.

This simplification may have reduced the accuracy in modeling the distinct shadowing characteristics of each environment. Specifically, the simulation might underestimate the extent of signal blockage in the downtown area and overestimate it in the suburban setting.

Future work could explore the impact of varying building height distributions to enhance the realism of the results.

APPENDIX: PYTHON FUNCTION

The following Python function is used to compute the PRR for each vehicle and the global PRR in the scenario. The input parameters to the function are pandas' DataFrames: tx and rx, representing the total number of packets transmitted and received by each vehicle.

REFERENCES

[1] OMNeT++ Community, "How to calculate network packet delivery ratio in OMNeT++," *OMNeT++ Manual*. [Online]. Available: <https://omnet-manual.com/how-to-calculate-network-packet-delivery-ratio-in-omnet/>.

```
def get_prr_global(tx,rx):
    prr={}
    # These are Series with carid as index
    tx_series = tx['value']
    rx_series = rx['value']
    car_ids = tx_series.index.union(rx_series.index)

    total_tx = tx_series.sum()
    prr_data = []
    for car in rx_series.index:
        tx_by_this_car = tx_series.get(car, 0)
        tx_by_others = total_tx - tx_by_this_car
        rx_by_this_car = rx_series[car]
        # Check if tx_by_others is a Series and if so, use its
sum
        if isinstance(tx_by_others, pd.Series):
            tx_by_others = tx_by_others.sum()

        if tx_by_others > 0:
            prr[car] = rx_by_this_car / tx_by_others
        else:
            prr[car] = 0

    prr_df = pd.DataFrame.from_dict(prr, orient='index',
columns=['prr'])
    prr_df.index.name = 'carid' # Set index name
    prr_df = prr_df.sort_index()

    prr_global = sum(prr.values()) / len(prr)
    return prr_df,prr_global
```